Haoyu Yang

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Employment

Harvard T.H. Chan School of Public Health *Boston, USA*

Postdoctoral Research Fellow, Department of Biostatistics · Advisor: Professor Xihong Lin Jul. 2023 – present

Education

Renmin University of China *Beijing, China*

Ph.D. in Statistics · Advisor: Professor Yang Li Sep. 2018 – Jun. 2023

Tianjin University *Tianjin, China*

B.S. in Mathematics and Applied Mathematics; minor in Financial Management Sep. 2014 – Jun. 2018

Research Interests

Methods: Synthetic data and generative modeling; AI-imputed phenotypes; robust and transfer inference; large-scale mediation analysis and composite-null testing; experimental design and covariate-adaptive randomization

Applications: Genome-wide and proteome-wide association studies in biobanks; epigenomics; clinical trials; economic and network experiments

Publications

Peer-reviewed journal articles (published or accepted):

[1] **Yang, H.**, Wang, R.†, Lin, Y., Lin, X.* (2026) A tail likelihood ratio method for large-scale causal mediation tests in epigenome-wide studies. *Journal of the American Statistical Association*, accepted.

[2] **Yang, H.**, Qin, Y., Yu, Y., Li, Y.* (2026) Design strategies for networked experiments via interference balancing. *Journal of Business & Economic Statistics*, published online, 1–14. doi:10.1080/07350015.2026.2658067

[3] Zhou, H., Li, Z., Shyr, D., Li, X., **Yang, H.**, Dey, R., …, Lin, X. (2026) Comparison of variant callers using 60,532 multi-ancestry whole genome sequences. *Briefings in Bioinformatics*, 27(2): bbag130.

[4] **Yang, H.**, Liu, Z., Wang, R., Lai, E., Schwartz, J., Baccarelli, A., Huang, Y., Lin, X.* (2025) Causal mediation analysis for integrating exposure, genomic, and phenotype data. *Annual Review of Statistics and Its Application*, 12: 337–360.

[5] **Yang, H.**, Qin, Y.†, Li, Y.*, Hu, F. (2024) Sequential covariate-adjusted randomization via hierarchically minimizing Mahalanobis distance and marginal imbalance. *Biometrics*, 80(2): ujae047.

[6] Qin, Y., Li, Y., Ma, W., **Yang, H.**, Hu, F.* (2024) Adaptive randomization via Mahalanobis distance. *Statistica Sinica*, 34(1): 353–375.

[7] Li, Y., Xie, J., Qin, Y., **Yang, H.*** (2023) Network-adjusted response-adaptive randomization in economic experiments (in Chinese). *Statistical Research*, 40(10): 151–160.

[8] Li, Y., **Yang, H.**, Yu, H., Huang, H., Shen, Y.* (2023) Penalized estimating equations for generalized linear models with multiple imputation. *The Annals of Applied Statistics*, 17(3): 2345–2363.

[9] **Yang, H.**, Qin, Y.†, Wang, F., Li, Y.*, Hu, F. (2023) Balancing covariates in multi-arm trials via adaptive randomization. *Computational Statistics & Data Analysis*, 179: 107642.

[10] Li, Y., **Yang, H.**, Yu, H., Huang, H., Shen, Y.* (2023) Penalized weighted least-squares estimate for variable selection on correlated multiply imputed data. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 72(3): 703–717.

[11] **Yang, H.**, Qin, Y., Li, Y.* (2022) Group sequential randomization for split questionnaire design (in Chinese). *Systems Science and Mathematical Sciences*, 42(1): 17–34.

Manuscripts in revision or under review:

[12] **Yang, H.**†, Wang, R.†, Song, S., Lin, X.* Synthetic phenotype assisted linear mixed models improve proteome-wide genetic discovery in incomplete biobank data. *Nature Communications*, major revision.

[13] **Yang, H.**, Qin, Y., Shi, C., Ren, W., Li, Y.* Adaptive weighted random isolation (AWRI): a simple design to estimate causal effects under network interference. *INFORMS Journal on Computing*, major revision.

[14] Fang, L., Qin, Y., Li, Y., **Yang, H.*** Sequential covariate-adjusted randomization framework. *Statistica Sinica*, minor revision.

[15] **Yang, H.**, Qin, Y., Li, Y.* Estimation of the average treatment and spillover effects in observational studies under general interference. Under review.

Manuscripts in preparation:

[16] **Yang, H.**, Wang, R., Lin, X.* Heterogeneity-aware empowerment via adaptive robust transfer for target-population inference (HEART). In preparation for submission.

** Corresponding author; † co-first authors (equal contribution).*

Book chapters:

- Variable selection for missing data. Chapter in *Modern Survey Analysis*.
- Statistical principles of data science. Chapter in *Introduction to Data Science*.

Software (R packages, github.com/haoyu-yang001):

- **TLR**: tail likelihood ratio and M-DACT tests for large-scale causal mediation under a composite null, with empirical-null corrections, competing tests (Sobel, DACT, HDMT) and simulation tools. Accompanies [1].
- **SynPALM**: synthetic phenotype assisted linear mixed models for GWAS with partially observed, AI-imputed phenotypes; sparse-GRM mixed models and chunked genome-wide score tests. Release v0.1.1 archived at doi:10.5281/zenodo.21782545. Accompanies [12].
- **HEART**: implements HEART [16] for target-population association testing; individual-level, doubly robust and summary-statistic modes with FDR/FWER control; online documentation.
- **CARM** (CRAN, co-author): covariate-adjusted adaptive randomization via Mahalanobis distance for two-arm, multi-arm and stratified trials. Accompanies [6] and [9].
- **SCR**: sequential covariate-adjusted randomization by hierarchical minimization of Mahalanobis and marginal imbalance; one-at-a-time allocation, imbalance measures, randomization tests and a multi-arm version. Accompanies [5].

Data resource:

- **Syn-PALM results portal** (syn-palm.genohub.org): genome-wide summary statistics for 2,916 plasma proteins in 402,086 UK Biobank participants.

Presentations

Statistical Genetics Meeting 2025; ILCCO Biostatistics Meeting 2026; JSM 2026 — *Boston, USA*
Robust synthetic data–assisted genome-wide association studies boost proteome-wide genetic discovery with partially observed data in biobanks — Nov. 2025; Mar. 2026; Aug. 2026

Joint Statistical Meetings — *Portland, USA*
Tail likelihood ratio method for large-scale causal mediation hypothesis testing — Aug. 2024

Statistical Genetics Meeting — *Boston, USA*
Causal mediation analysis for integrating exposure, genomic, and phenotype data — Jun. 2024

The 37th New England Statistics Symposium — *Connecticut, USA*
Design strategies for networked experiments via interference balancing — May 2024

The Ph.D. Forum on Biostatistical Methods and Applications — *Beijing, China*
Adaptive randomization via Mahalanobis distance — Jun. 2021

Awards and Honors

Jun. 2026	Poster Award, 2026 ICSA Applied Statistics Symposium
Oct. 2022	Top Ten Papers, 8th National Postgraduate Statistics Forum
Aug. 2021	Top Ten Papers, 2nd Data Science and Modern Economic Statistics Forum
Dec. 2020	First Prize, “BeiGene” Excellent Paper for Youth

Teaching and Mentoring

Teaching assistant, Renmin University of China:

2021	Probability Theory
2021	Complex Statistical Analysis
2020	Introduction to Data Science

Research mentoring, Harvard:

Current	Research mentor to a Harvard College junior on an undergraduate research project
Current	Research mentor to a Ph.D. student in Professor Xihong Lin’s group

Applied Research Projects

Research Associate, during Ph.D. studies:

Gas-station selection for marketing and evaluation of oil marketing effects (with CPPEI) *2022*

- Developed discount strategies for decision makers and an evaluation process for the performance of each gas station. Estimated individual treatment effects from observational and experimental data with geographic POI information, using causal inference and machine learning methods.

Responsible consumption: cognition, behavior and influencing factors (with JD.com) *2020*

- Studied Chinese consumers' cognition, behavior and attitudes toward responsible consumption. Built a Responsible Consumption Index (RCI) through questionnaire design and survey, principal component analysis and factor analysis.

Factor analysis of active and healthy aging *2020*

- Constructed an index system for active and healthy aging and analyzed the health of older adults from community-level data, using missing-data imputation, descriptive statistics, cluster, contingency and correlation analyses, and hypothesis tests.

Geographic information analysis and sales forecasting for gas stations (with CPPEI) *2019*

- Built a rating model for gas stations from geographic POI information, and forecast oil and non-oil sales from operating and spatial data using factor analysis, clustering and classification methods.

Professional Service

- Journal referee: *Statistica Sinica*; *Biostatistics*; *Journal of Data Science*.

References

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